Dockerized Sports Planning Assistant Service Using Python

Report

Course Name: Deveops

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	Kartik Dubey	EN22CS301486
2.	Jyoti Singh Yadav	EN22CS301473
3.	Kapil Daunde	EN22CS301482
4.	Madhur Bhawsar	EN22CS301560
5.	Jaspreet Kour Rathod	EN22CS301456
6.	Kunal Patidar	EN22CS301537

Group Name: 10D5

Project Number:DO-10

Industry Mentor Name: AASHRUTI SHAH MAAM

University Mentor Name:PROF.RITESH JOSHI SIR

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Introduction:

This system is designed to provide **sports-related backend services** such as match scores, player statistics, and game data through REST APIs.

The application is built using Python and follows a **microservices-based architecture**, where each service is independently developed and deployed inside Docker containers.

For cloud infrastructure, the system uses **Amazon Web Services**, specifically Linux-based virtual machines (EC2 instances).

To automate the development lifecycle (build, test, deploy), a CI/CD pipeline is implemented using **GitHub Actions**, ensuring fast and reliable deployments.

Project Report

Dockerized Sports Planning Assistant

1. Problem Statement & Objectives

Problem Statement

Modern sports organizations deal with large volumes of data such as match statistics, player performance, and live updates. However, traditional systems often suffer from:

- Data silos and lack of integration
- Deployment inconsistencies ("it works on my machine" issue)
- Limited intelligence (only static data retrieval)
- Difficulty in scaling and maintaining systems

These challenges make it difficult to build reliable and intelligent sports applications.

Project Objectives

The main objectives of this project are:

- To develop a **Python-based backend system** for sports data processing
- To implement **containerization using Docker** for consistent environments
- To build a **Multi-Agent AI system** capable of reasoning and generating insights
- To automate deployment using CI/CD with **GitHub Actions**
- To deploy the system on **Amazon Web Services** cloud infrastructure
- To create a scalable and production-ready architecture

2. Scope of the Project

The scope of this project includes:

- Development of an API-driven backend system
- Integration of AI agents for real-time sports analysis
- Deployment on AWS EC2 (Linux-based instances)
- Implementation of CI/CD pipeline for automation
- Creation of an interactive frontend dashboard

The project goes beyond a simple database system by introducing an **AI-powered assistant** that can analyze and generate sports insights dynamically.

3. Proposed Solution

To overcome the limitations of traditional systems, we designed a **Dockerized,**

Multi-Agent AI architecture.

Key aspects of the solution:

- Use of **Docker containers** to ensure consistent deployment

- Implementation of **AI agents** using CrewAI for intelligent decision-making
- Integration with external APIs to fetch live sports data
- Deployment on AWS for scalability and reliability
- Automation using CI/CD pipelines

This approach transforms the system from a static backend into an **intelligent sports assistant** capable of reasoning and analysis.

4. Key Features

- Multi-Agent AI system (Researcher + Writer agents)
- Real-time sports data retrieval
- Automated CI/CD pipeline
- Dockerized microservices architecture
- Cloud deployment on AWS
- Interactive dashboard for user interaction
- Scalable and fault-tolerant design

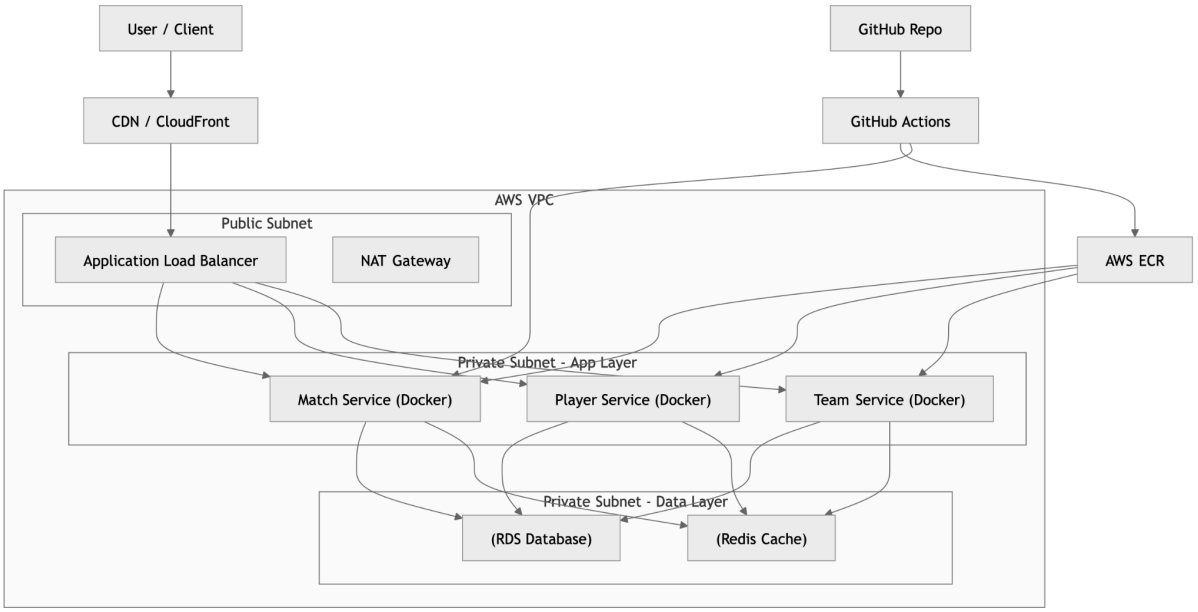
5. Overall Architecture / Workflow

The system follows a modern DevOps-based workflow:

1. Developer pushes code to GitHub
2. CI/CD pipeline is triggered via **GitHub Actions**
3. Application is built and tested automatically
4. Docker image is created
5. Image is pushed to AWS ECR (container registry)
6. AWS EC2 instance pulls the latest image
7. Application is deployed and runs inside Docker containers

Additionally:

- AI agents process sports queries
- External APIs provide real-time data
- Results are displayed through a frontend dashboard



6. Tools & Technologies Used

Frontend

- Streamlit (Interactive dashboard)

Backend

- Python
- CrewAI (Multi-agent orchestration)
- LangChain

AI & Intelligence

- Google Gemini API (LLM for reasoning and analysis)

DevOps

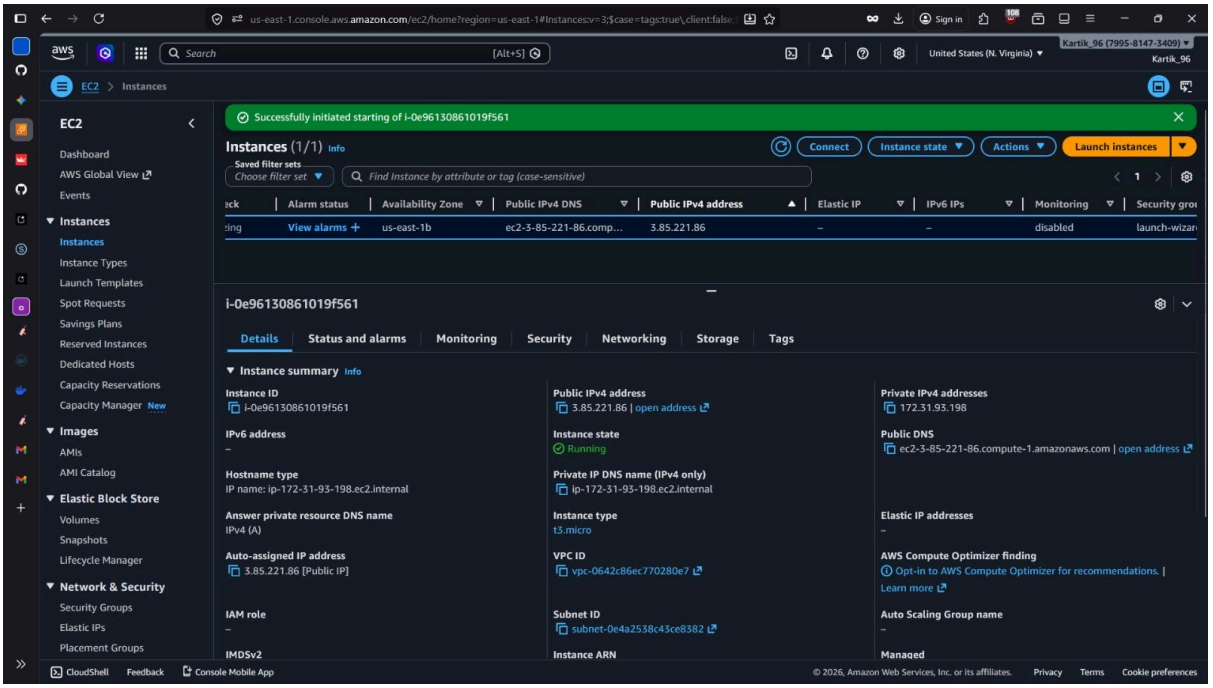
- Docker (Containerization)
- **GitHub Actions** (CI/CD automation)

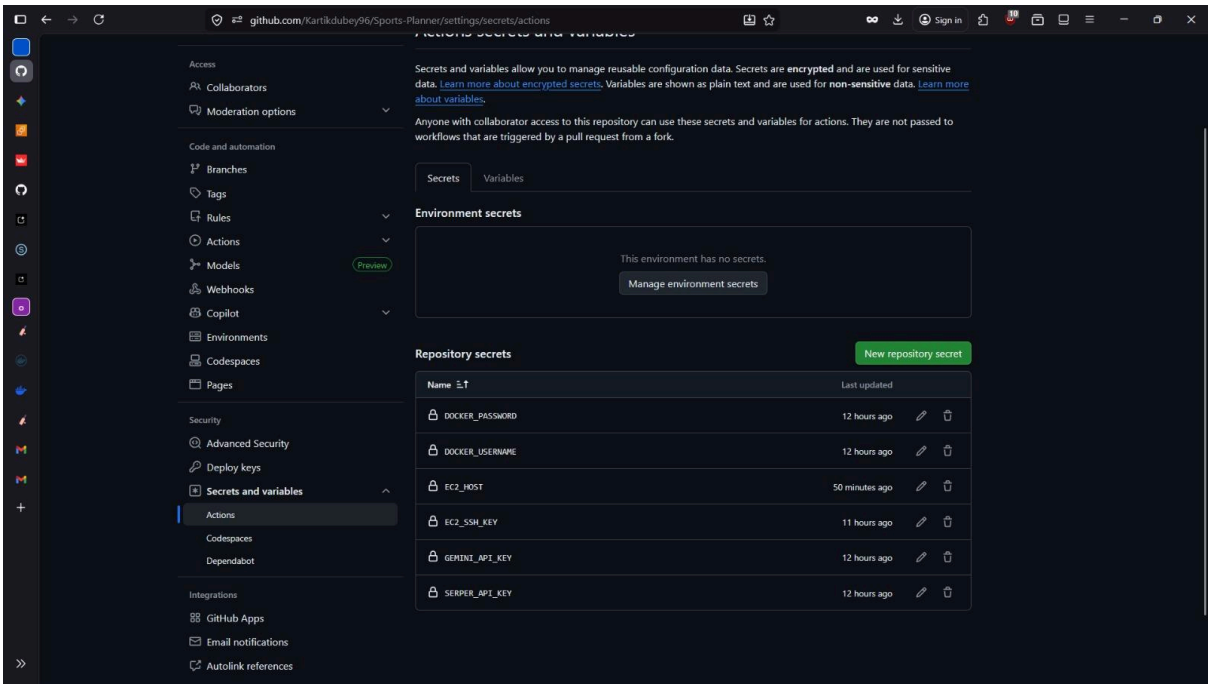
Cloud

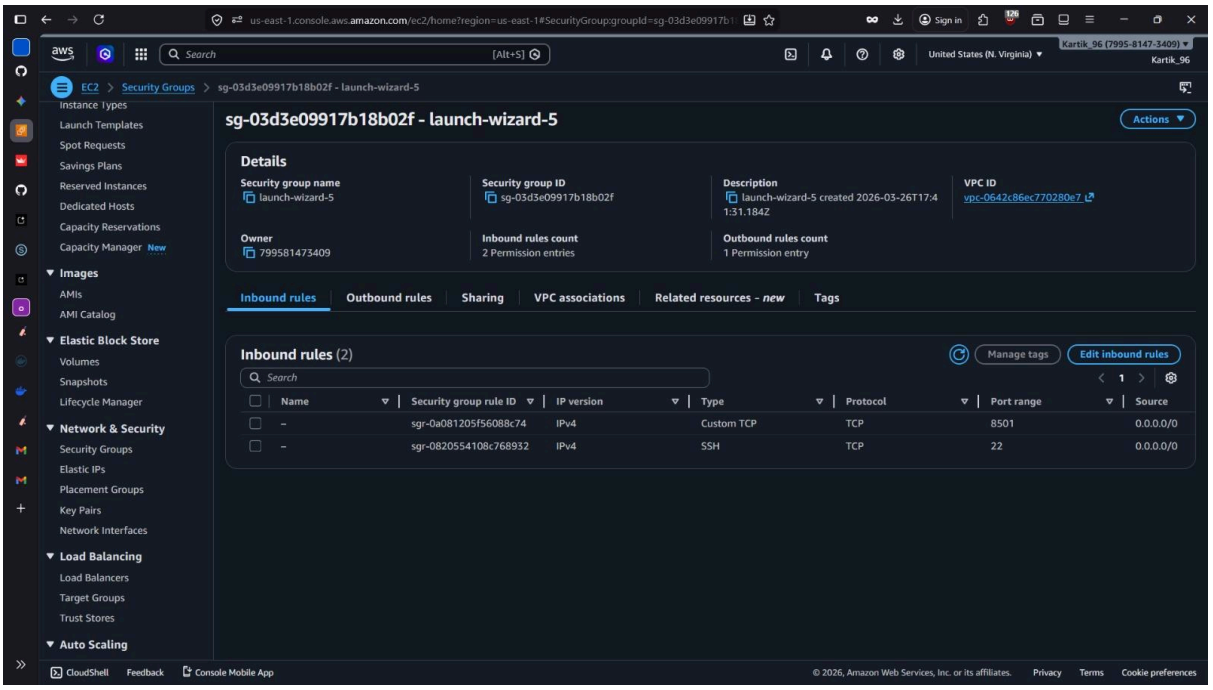
- **Amazon Web Services** EC2 (Linux OS)
- AWS ECR (Container registry)

7. Results & Output

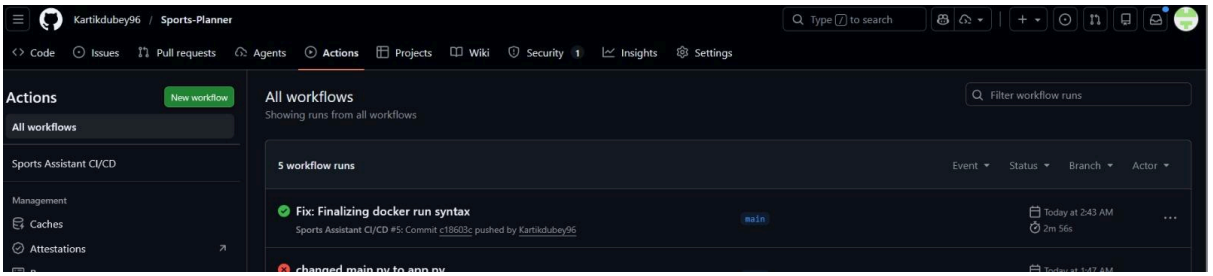
Screenshots / Outputs

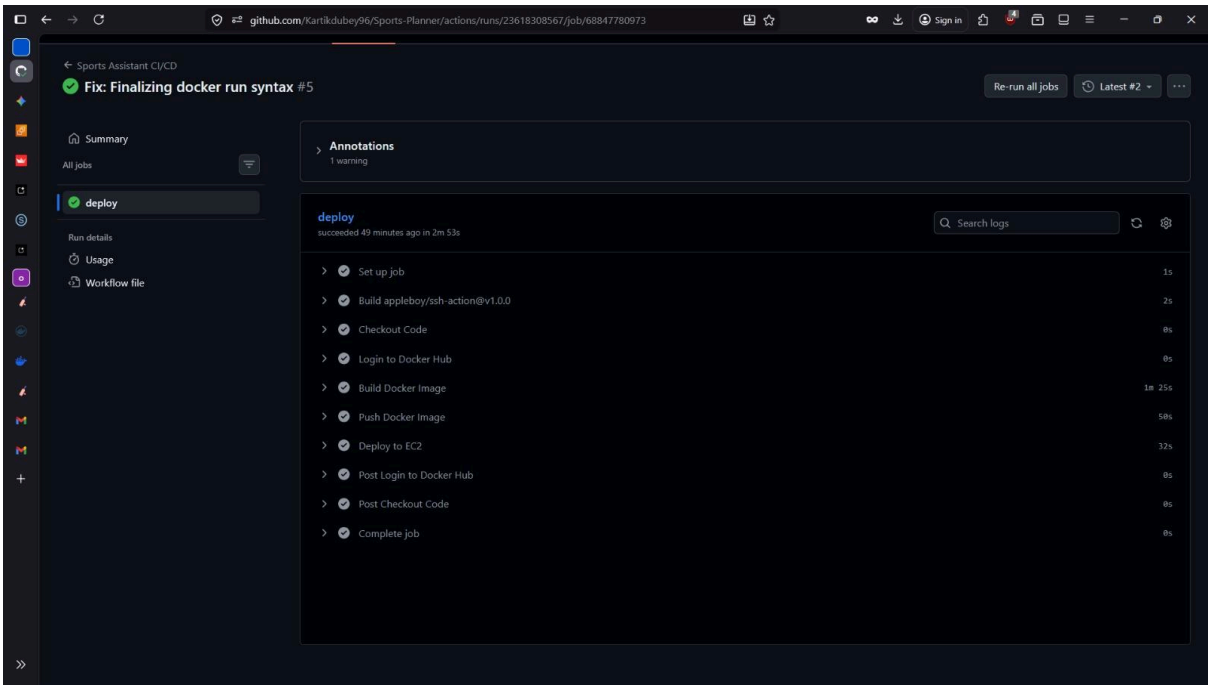


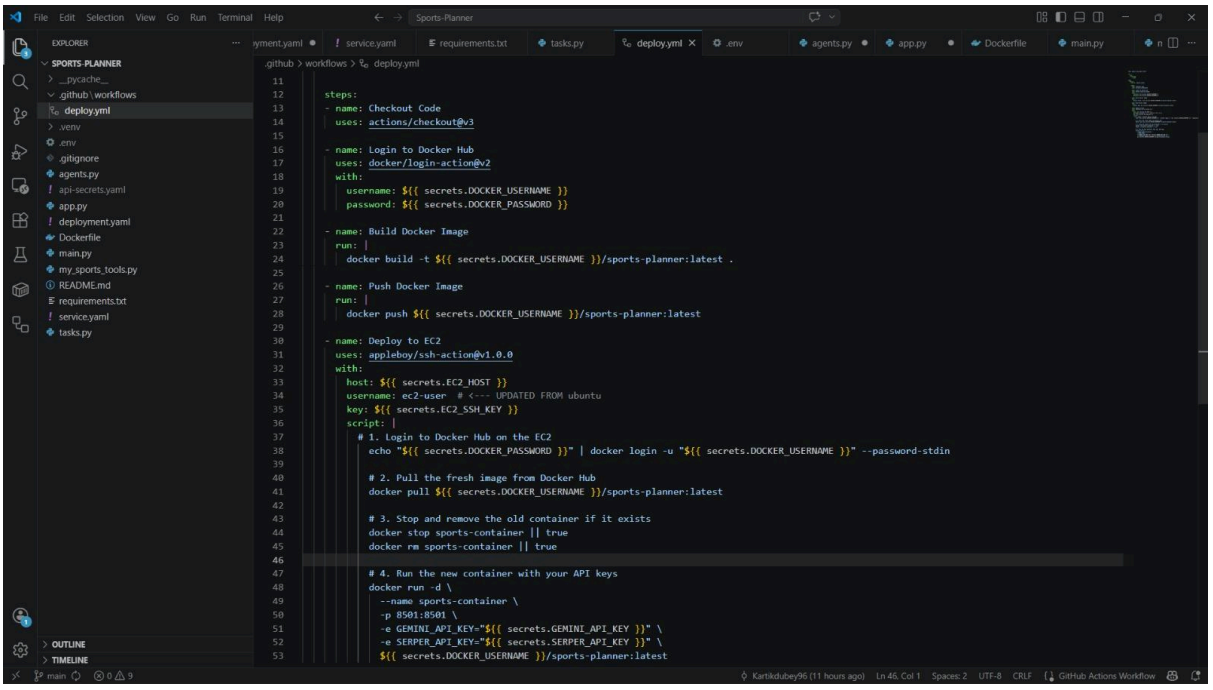




- GitHub Actions pipeline showing successful (green) builds
- Running application hosted on AWS EC2 (via Public IP)
- Streamlit dashboard displaying AI-generated sports reports

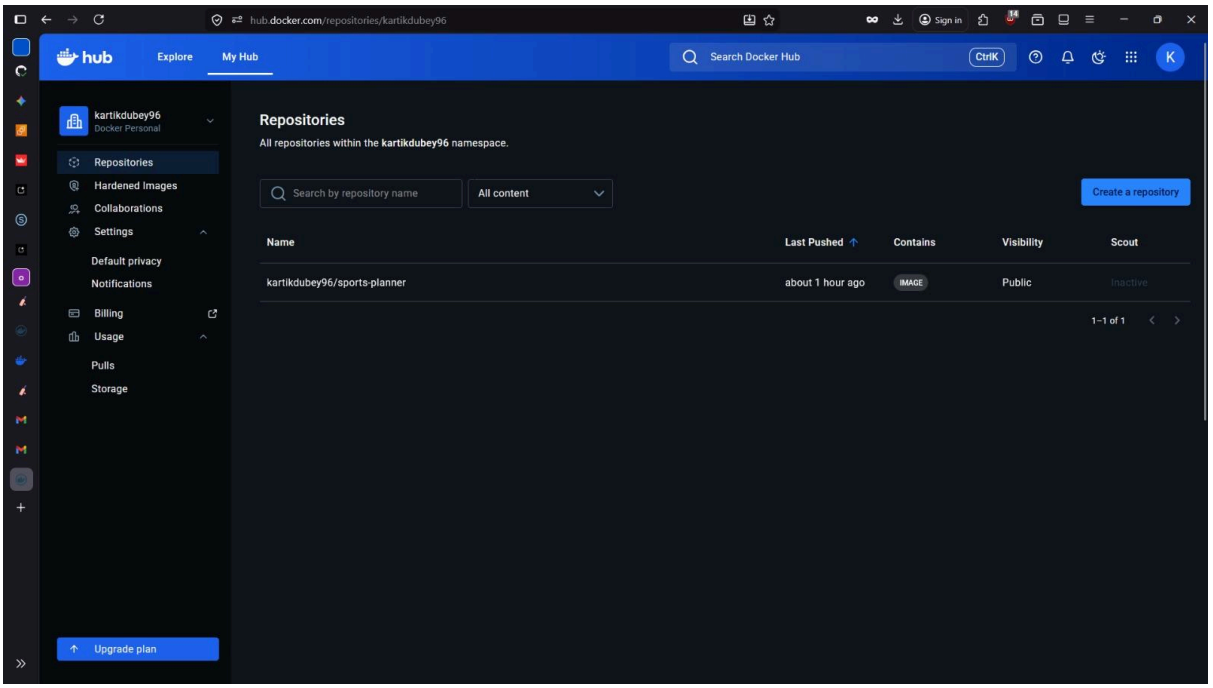


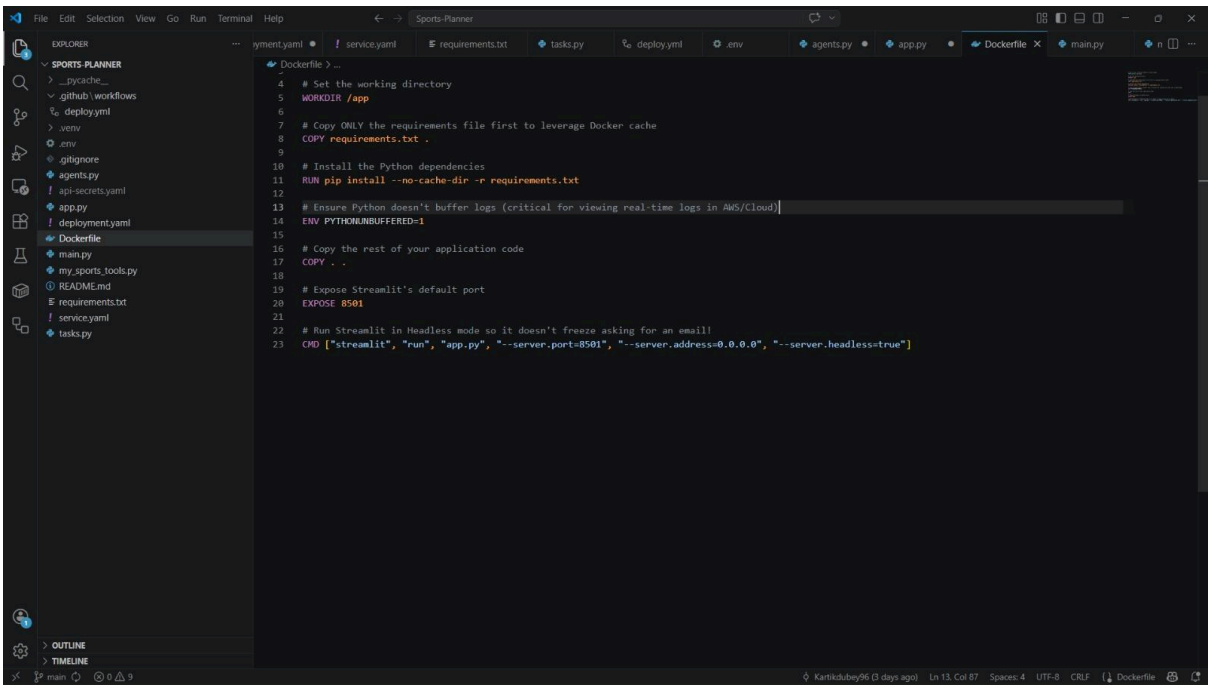




Reports / Dashboards / Models

- AI-generated sports analysis reports
- Interactive dashboard for user queries
- Real-time insights generated using LLM

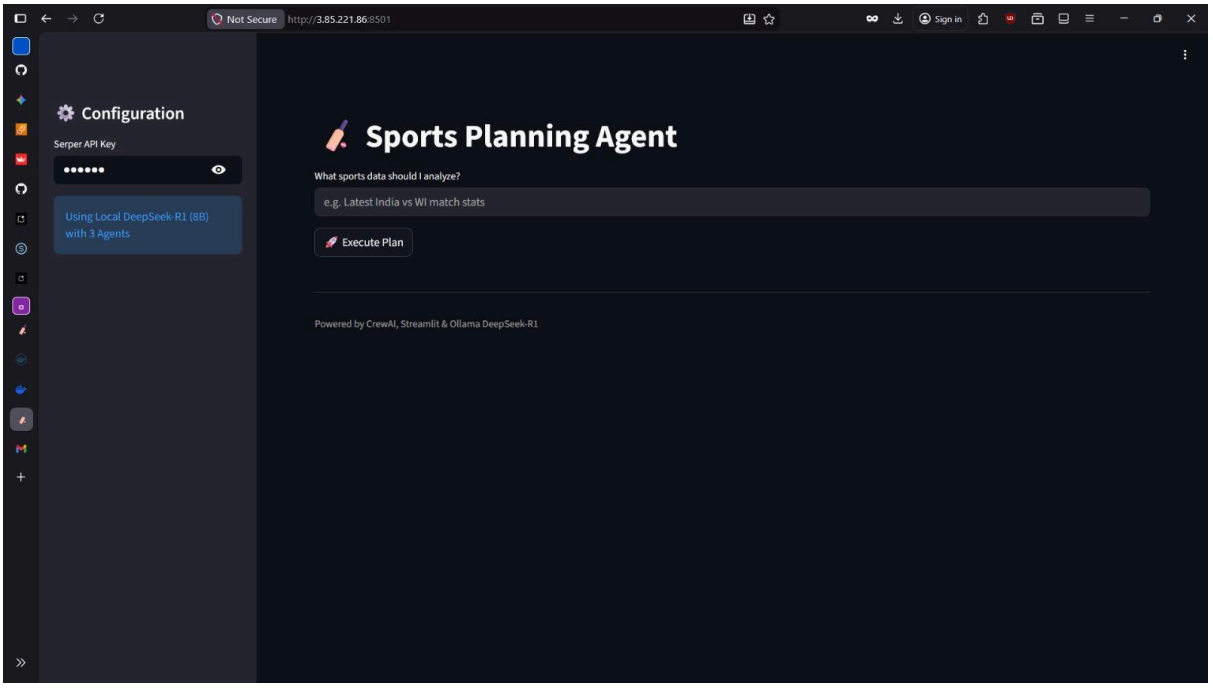




Key Outcomes

- Fully automated CI/CD pipeline

- Successful deployment on AWS cloud
- Real-time AI-based sports assistant
- Elimination of environment inconsistency issues
- Faster and reliable deployments



8. Conclusion

This project successfully demonstrates how combining **AI-based multi-agent systems** with **modern DevOps practices** can create a powerful and scalable application.

By using Docker for containerization and CI/CD pipelines for automation, we eliminated deployment inconsistencies and improved system reliability. The integration of AI agents enables the system to go beyond simple data retrieval and provide intelligent insights.

Overall, the project highlights the importance of **automation, scalability, and intelligent systems** in modern application development.

9. Future Scope & Enhancements

- Integration with more sports data APIs
- Implementation of Kubernetes (EKS) for container orchestration
- Advanced analytics and predictive modeling
- Enhanced UI/UX for better user experience
- Real-time notifications and alerts
- Improved security using advanced IAM roles and monitoring tools
- Integration with mobile applications